INSTITUTO POLITÉCNICO NACIONAL
ESCUELA SUPERIOR DE CÓMPUTO

Introduction to Cryptography

Lab 07: Tiny Block Cipher *March 26, 2026*

Please do the following exercises by your own using C/C++ programming language and bitwise operators. The evaluation of these exercises is as follows:

1. Key expansion

Implement a function key_expansion that receives as parameter an unsigned short int variable $K = k_0k_1 \cdots k_{15}$ and an sbox S, where k_i is a bit. Then split K in two words $w_0 = k_0k_1 \ldots k_7$ and $w_1 = k_8k_9 \ldots k_{15}$ i.e. w_0 and w_1 has 8 bits and compute the following values:

1. $w_2 = w_0 \oplus 10000000 \oplus S(r(w_1))$
2. $w_3 = w_2 \oplus w_1$
3. $w_4 = w_2 \oplus 00110000 \oplus S(r(w_3))$
4. $w_5 = w_4 \oplus w_3$

If w is a word of 8 bits, then split in two blocks of 4 bits i.e. $w = n_0n_1$, then $r(w) = r(n_0n_1) = n_1n_0$. For example if $w = 11110000$ then $r(w) = 00001111$. Use your code of the previous lab session to generate an $S : \{0,1\}^8 \to \{0,1\}^8$. Print in hexadecimal the values $w_0, w_1, \ldots w_5$.

2. Programming the tiny block cipher

In this section you will implement a first version of a tiny block cipher with the following parameters The block size will be $|M| = 16$ bits; the key size will be $|K| = 16$ bits, and the number of rounds $r = 3$. The structure of the tiny block cipher is the following is shown in Algoritmo 1. K_0 will be the original key i.e. $K_0 = K$, K_1 and $K_2 = w_2w_3$ and $K_3 = w_4w_5$ and are constructed with the results obtained by the function key_expansion that you implemented before. S uses an sbox to substitute each block of 8 bits in M.

With this information do the following:

Algoritmo 1: Tiny encryption algorithm

Input : M a block of plaintext of 16 bits, $K = k_0k_1$ a pseudorandom key, where $|k_i|$=8 bits
Output: C a block of ciphertext

1 **for** $i \leftarrow 0$ **to** 2 **do**
2 $\quad M \leftarrow M \oplus K_i$
3 $\quad M \leftarrow S(M)$ `//substitution`
4 **end**
5 $C \leftarrow M$
6 **return** C

1. Design a computer program to generate a secret key K and to create a function S. Your program must fulfill the following requirements:
   a) Use a variable of 16 bits (unsigned short int) to store the key K. The key must be randomly generated. Store the key in hexadecimal in a text file.
   b) Use the function you implemented in the previous session to randomly generate an $S : \{0,1\}^8 \rightarrow \{0,1\}^8$. Store the generated table for S in a text file. Use hexadecimal to store it.
2. Design another computer program to encipher using the **tiny encryption algorithm**. Your program must fulfill the following parameters.
   a) Your program must ask the user the filename where the key is stored and the filename where the function S^{-1} is stored.
   b) Also your program must ask the user the plaintext $\mathcal{M}$ to be enciphered, this plaintext must be a string of any 2 printable ascii characters.
   c) Extract blocks of 8 bits from the given key K using bitwise operations to do operation of line 2 (in the pseudocode).
   d) To do operation in line 3, use the function you implemented in the previous session to substitute each block of 8 bits in M with the 8 bits given by the function S that you stored in a textfile.
   e) Print C in hexadecimal.

3. Products

Every student must write her/his own document to present results. Your document must include

1. Personal information, date of the lab session and the topic that we are studying in this lab session.
2. Source code of your function key_expansion
3. Screenshots of your source code running for section 1 and section 2.